

Indian Institute of Technology Palakkad

Department of Mathematics

MA 5015 (Numerical analysis with programming)

2026 (January-May)

Instructor: Dr Gopikrishnan C R

Teaching assistant: Ms Alphonsa George

Office: 10, D03-FF (Dr. APJ Abdul Kalam Block).

Schedule (Test I)

Regular classes and tutorials (T,Th) - N308, 12.00 - 12.50 PM

Regular labs (W) - A01-201, 9.00-11.50 AM

Colour codes follow as lectures (yellow), and labs (blue).

Date	Lec/Lab/Tut	Topic
January 7	Lab 1	Elementary unix commands, file management in unix, Emacs and Vim, python data types
January 8	Lecture 1	Finite digit arithmetic
January 13	Lecture 2	Error propagation
January 14	Lab 2	List management and control structures
January 15	Lecture 3	Asymptotic convergence
January 20	Lecture 4	Bisection method
January 21	Lab 3	Functions in python, type conversion, local and global variables, Jupyter notebook
January 22	Lecture 5	Newton method
January 27	Lecture 6	Secant method
January 28	Lab 4	Implementation of Newton and Secant methods
January 29	Lecture 7	Fixed point method
February 3	Lecture 8	Matrix norms
February 4	Lab 5	Implementation of fixed point methods, computation of matrix norms
February 5	Lecture 9	Gaussian elimination
February 10	Lecture 10	LU and QR algorithms

February 11	Lab 6	cmath, Numpy, Scipy, Implementation of Gaussian and LU method
Test I syllabus		